Team Capability — PROJECT Scenario

(Armoured Battle Console)

a. Scenario

Note: The ALL TEAM must demonstrate the Armoured Battle Drill (Terrain-Aware Armoured Engagement & High-Value Target Neutralisation) on **AI**-DSSW on at least one (1) domain/theatre of operations for the PROJECT, with the same capabilities extensible to additional domains post-PROJECT.

A BLUE armoured battle group, commanded by a human player, deploys on real-geography 3D terrain in the India–Pakistan border region to locate and destroy a RED armoured force's (OPFOR) High-Value Target (HVT) while protecting its own. RED is controlled either by an **AI** Red player or a second human player. (For PROJECT purposes, the RED player must be **AI**-controlled / rule-based.)

The BLUE element deploys in tactical formations, advances across the terrain, detects and engages the RED force, and destroys the RED HVT while protecting its own. The battle is resolved by a tick-based simulation engine that accounts for terrain elevation and slope, line of sight, sensor ranges, vehicle mobility, fuel, and realistic terrain-aware ballistics and hit-probability modelling. Every tick is recorded in full for live rewind, complete replay and post-battle analytics.

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b. Players

- **Exercise Controller with AI Capabilities (Administrator):** Creates the scenario and sets its duration; ingests the terrain data (for PROJECT purposes, any open-source 3D globe engine that shows terrain may be used, so no separate terrain file needs to be sourced or uploaded in advance); draws the BLUE and RED deployment zones and sets the allowed inventory for each force; chooses whether RED is a human or an **AI** Red player and generates the force keys; watches the battle live, then reviews the replay and results.
- **BLUE Element (human player, joining with the BLUE force key):**
  - **Formation ALPHA (Assault):** A tank-heavy strike formation responsible for closing with the RED force and driving on the RED HVT.
  - **Formation BRAVO (HVT Defence):** A defensive formation responsible for holding a ring around the BLUE HVT and intercepting intruders.
  - **Formation CHARLIE (Fire Support):** An artillery formation responsible for indirect area fire on detected RED clusters.
  - **Formation DELTA (Reconnaissance):** A fast, long-range sensor formation responsible for discovering the RED HVT and advance-warning of RED movement.
  - Formation composition (unit types and counts) is configurable by the player, within the inventory the administrator allows.
  - **AI formation suggestion:** The BLUE player may request an **AI**-generated recommendation for formation composition, shape and doctrine, proposed by the **AI** Tactical Advisor from the declared inventory, the drawn zones and the terrain; the player may accept, adjust or disregard the suggestion entirely, with command authority remaining with the human player throughout.
- **RED Element (Opposing Force / OPFOR):** (For PROJECT purposes, the RED player must be **AI**-controlled / rule-based.)
  - **AI Red player:** Runs fully autonomously on **AI** / rule-based logic, with no human intervention required at any point once the battle starts. RED's order of battle is generated automatically to a default composition or to whatever inventory the administrator sets — and placed at random inside the RED zone, with each unit given a doctrine and an avenue of approach to advance along. From force generation through detection, movement and engagement, every decision RED makes — who to engage, when to hold, flank, advance or fall back is computed by the engine itself, so the RED side can run an entire battle unattended.
  - **RED Human (future capability):** A second player joins with the RED force key and uses the same deployment and command tools as BLUE along with **AI** autoplay option controlled by the admin.

c. Objective

- **Blue Team Objective:** Locate the RED HVT and destroy it, while keeping the BLUE HVT intact, before the time limit. Alternatively, destroy every RED unit.
- **Red Team Objective:** Protect the RED HVT and intercept any BLUE units that enter the RED zone. Destroy the BLUE HVT or eliminate the BLUE force.
- **End conditions (the battle ends automatically when any one is met):** either HVT is destroyed; all BLUE units are destroyed; all RED units are destroyed; or the time limit is reached.
- **Winner determination:** Destroying the enemy HVT decides the winner outright and overrides every other measure. If neither HVT is destroyed, the side with more confirmed kills wins; equal kills is a DRAW. If both HVTs are destroyed, the battle is declared a STALEMATE.

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d. Terrain & Environmental Parameters

- **Environment:** Any open-source, real-geography 3D globe engine with terrain support, rendering satellite or aerial imagery, national border overlays, and a numbered coordinate grid, with a camera that opens framed on the area of interest. (For PROJECT purposes, any open-source 3D globe engine that shows terrain may be used.)
- **Area of Operations:** Two polygon deployment zones, BLUE and RED, drawn directly on the map by the Exercise Controller — either by clicking vertices or by a freeform lasso — each erasable and redrawable.
- **Terrain Model:** Real elevation data is ingested and processed into a fine grid that derives, per cell, elevation, slope, a terrain classification (flat / sloped / steep), a mobility score and a concealment score. Vehicles move and turn according to the actual ground slope beneath them, slowing on steeper ground and unable to climb slopes beyond what a given platform can handle. Where no terrain data is ingested for an area, the engine falls back to flat, unobstructed going so the drill can still run. (For PROJECT purposes, no separate terrain file needs to be sourced or uploaded — any open-source 3D globe engine that already shows terrain may be used directly.)
- **Terrain-derived mobility:** Movement speed and feasibility are shaped by slope and each platform's climbing ability, with automatically generated mobility corridors (avenues of approach) between the two zones that give faster, road-like movement to units travelling near them, and slower off-road movement elsewhere.
- **Concealment & Intervisibility:** Terrain concealment reduces how far an enemy can detect a unit and how easily it can be hit. Line of sight between any two points is computed live against the terrain surface, including a correction for the curvature of the earth over long distances.

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e. Blue Team Plan

1. **Join & Deployment:** Enter the BLUE force key, build formations by name, doctrine, shape and composition, and deploy them facing the RED zone.
2. **Objective Placement & Ready:** Place the BLUE HVT inside the BLUE zone and mark the force Ready; the battle starts once both forces are ready.
3. **Advance & Route Plotting:** Select a formation, plot a waypoint route on the map, set its doctrine, and advance the battle tick by tick or continuously on Auto.
4. **Reconnaissance & Contact:** DELTA's long-range sensors detect RED units within range and line of sight, and reveal the RED HVT once it is discovered.
5. **Engagement & Mid-Battle Command:** ALPHA engages with direct fire and CHARLIE fires indirect area fire; any formation or individual unit can be re-tasked live — including clicking a unit to redirect its path mid-simulation — while the player monitors through the unit list, intrusion alerts and the spectator camera.

6. **HVT Neutralisation & Review:** Sustained fire from units in range and line of sight reduces the RED HVT to zero health for a BLUE win; the results pop-up, full dashboard and replay follow automatically, with Play Again to restart from the same opening positions.

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f. Red Team Plan (AI Agent Red Player)

1. **Force Generation & Posture:** RED units are generated and controlled by the **AI** Red agent: it spawns its own order of battle at random inside the RED zone, independently assigns each unit a weighted doctrine (Aggressive / Flanking / Suppressive / Defensive) and an avenue of approach, with no human input required at this or any later stage.
2. **Detection & Home Defence:** The **AI** Red agent independently builds its own detection picture each tick from its sensors and line of sight, and autonomously dispatches nearby units to intercept any BLUE units detected inside its zone — a decision made and executed by the **AI** without administrator or player intervention.
3. **Engagement & Counter-Attack:** The **AI** Red agent selects its behaviour according to doctrine — hold and fire, advance to stand-off range, flank at an offset, or close to effective range — continuously re-evaluating this choice as the engagement develops, with Aggressive and Flanking units ultimately advancing on the BLUE HVT while Defensive and Suppressive units hold near the RED HVT.

*(RED Human is a future capability of the platform and outside PROJECT scope; for the PROJECT, RED runs as the **AI** Red player described above.)*

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g. Actions by Blue and Red Players

Phase	Blue Team Actions	Red Team Actions
Deployment	Deploys formations, places HVT, marks Ready.	**AI** Red player: auto-generated at start. Human: same deployment tools as BLUE.
Advance	Plots routes, issues orders, advances ticks; speed shaped by terrain and corridors.	Aggressive/Flanking advance on the BLUE HVT; Defensive/Suppressive hold near the RED HVT.
Detection	RECON detects RED units and the RED HVT within sensor range/line of sight.	Detects BLUE the same way; dispatches responders to intruders.
Engagement	Auto-selects best target/round per unit; support units fire indirect area fire.	Engages per doctrine; returns fire with the same combat model.
Battle End	HVT destroyed → victory banner, results dashboard, replay.	Destroying the BLUE HVT → RED victory.

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h. Live Simulation with AI Features

During an active engagement cycle, the simulation shall furnish observable, real-time confirmation of the following capabilities within the operator viewport:

- **Tick engine:** The engagement is governed end-to-end by a discrete, tick-based simulation loop, progressing incrementally or continuously at a configurable cadence, with every tick fully resolved, timestamped and persisted for subsequent reconstruction.
- **Live 3D unit models:** Every platform is rendered as a three-dimensional asset, correctly geo-positioned and orientated relative to the underlying terrain, with force affiliation immediately discernible to the operator.
- **Live health & status:** Each entity presents a continuously refreshed health and morale indicator superimposed on the map, alongside ordnance and fuel status telemetry, together with a discrete indicator denoting an immobilisation state.

- **Live firing visualisation:** Fire missions are rendered contemporaneously with the engagement, comprising tracer trajectories along the line of fire, muzzle discharge effects, and impact signatures (fireball, smoke, debris) upon a successful hit, with a corresponding dust signature denoting a miss — such that provenance and destination of every discharge remain unambiguously attributable.
- **Click-to-command:** Any formation may be interactively selected at any juncture during the live engagement to interrogate its current status and to effect a mid-simulation revision of its route or doctrine, without interrupting engine execution.
- **HVT tracking:** Both High-Value Targets are denoted by a distinct marker with continuously updated health and morale indicators once acquired within the opposing force's line of sight; upon neutralisation, the marker transitions to an unambiguous destroyed-state icon. An HVT not yet acquired by the opposing force remains withheld from that force's display until acquisition occurs.
- **Detection & fog of war:** An administrator-mediated control constrains each participant's operational picture to only those hostile entities currently within their own force's detection envelope, such that situational awareness derives from genuine reconnaissance rather than unrestricted visibility.
- **Intrusion & contact alerts:** A contemporaneous on-screen notification alerts a participant upon detection of hostile entities within their own designated zone, accompanied by a compositional breakdown of the incursive force.
- **Spectator camera:** A pursuit-view camera may be attached to any designated entity throughout the engagement, presenting its heading, operational status and immediate spatial context.
- **Rewind while live:** Antecedent ticks may be traversed and examined via a timeline control without arresting or otherwise perturbing the ongoing engagement, with resumption at the live boundary thereafter.

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i. AI Tactical Advisor

AI-DSSW is built around artificial intelligence as a decision-support layer, not merely as a simulation backend. A dedicated Tactical Advisor, powered by **AI Agent**, sits alongside the human commander throughout the engagement and continues to provide **AI**-generated insight after it ends:

- **Live AI threat assessment:** At a configurable tick interval, or on operator demand, the **AI** Tactical Advisor re-evaluates the requesting force's current detection picture and generates a concise, **AI**-authored threat assessment — identifying the most significant emerging risk, the formation it bears upon, and the tempo at which it is developing.
- **Ranked, AI-generated recommendations:** Alongside the threat assessment, the **AI** Tactical Advisor proposes three ranked courses of action, each with a stated **AI**-generated rationale grounded in the live tactical picture — for example, redirecting indirect fire onto a newly detected concentration, withdrawing an exposed flank, or accelerating an avenue of approach while a sensor gap persists. Recommendations are strictly advisory: the **AI** proposes, the human commander disposes, and no recommendation is executed automatically or without operator confirmation.
- **Ask the Battle — AI-generated after-action interrogation:** Once the engagement concludes, the same **AI** system exposes a natural-language question-and-answer interface over the persisted tick logs, kill feed and casualty record — so questions such as "Why did Formation ALPHA lose three tanks?" or "Which entity destroyed RED MBT-2?" receive an **AI**-generated answer traceable directly to the recorded data, rather than a speculative response.
- **AI-assisted formation suggestion (BLUE):** During deployment, the **AI** Tactical Advisor can propose a recommended formation composition, shape and doctrine for the BLUE player — generated from the declared inventory, the drawn zones and the terrain — which the player may accept, adjust or disregard entirely; command authority remains with the human player throughout.
- **AI-driven RED agent behaviour:** The RED side is itself an **AI** agent rather than a fixed script: it independently interprets its sensor picture, selects and re-selects a doctrine-consistent course of action tick by tick, and adapts its posture — holding, flanking, massing or withdrawing — in direct response to how the engagement unfolds, so that no two **AI**-run battles need play out identically even from the same starting position.

- **AI-generated scenario briefings:** Before an engagement begins, the **AI** Tactical Advisor reviews the configured zones, inventories and doctrines and generates a short, human-readable mission briefing for the Exercise Controller and players — summarising the relative force balance, notable terrain features and the likely axis of contact — so every participant starts from an **AI**-prepared situational picture rather than raw configuration data.

j. Platform & Weapon Inventory

A configurable platform and armament repository, amenable to modification by the end user absent any source-level intervention, thereby permitting the simulation to instantiate any desired order of battle.

- **Platform data:** Designation and unit taxonomy (MBT / IFV / APC / SPG / RECON); protective characteristics (armour thickness per facet); physical dimensions; mobility parameters (mass, motive power, on/off-road velocity, gradient tolerance, fuel capacity and consumption rate); firepower attributes (cyclic rate); and sensor characteristics (acquisition range, thermal-imaging capability, crew proficiency).
- **Weapon data:** Armament type and classification (kinetic-energy, chemical-energy, autocannon, or artillery-pattern), maximum effective range, precision/dispersion characteristics, and armour-defeat and area-effect damage parameters.
- **Governance:** The inventory is maintained under centralised administrative control and retained in engine cache; amendments take effect upon the subsequent initiation or reset of an engagement, thereby ensuring the reference data governing a live exercise remains internally consistent throughout.

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k. Scenario Authoring & Exercise Control (Administrator)

- **Command Panel:** Instantiate scenarios (designation, tick interval, temporal ceiling, descriptive metadata), observe a live status indicator and progression gauge per scenario, and administer, replay or purge them, with visibility restricted per administrator to the scenarios of their own authorship.
- **Tick Management:** The simulation progresses via discrete, uniform intervals designated ticks, with positional, detection, movement and combat state recalculated for every entity once per interval. The administrator prescribes the simulated-time duration attributable to a single tick at scenario creation, and the aggregate tick allocation (or the overall temporal ceiling) determines the maximum duration for which the engagement may proceed prior to automatic termination. A finer tick granularity yields smoother and more precise resolution of movement and engagement; a coarser granularity permits an equivalent step count to span a proportionally longer engagement.
- **Scenario Setup (guided, tabbed workflow):**
  - **Terrain:** Ingest terrain data for the chosen area of operations. (For PROJECT purposes, any open-source 3D globe engine that shows terrain may be used, so no separate terrain file needs to be sourced or uploaded in advance.)
  - **Zones:** Draw the BLUE and RED deployment zones on the map.
  - **Inventory:** Set the allowed roster per force, or leave it open for an unrestricted order of battle.
  - **Readiness:** A checklist confirming terrain, zones and both forces are ready before the battle can start.
  - **Force Keys:** Choose human or **AI** Red player for RED, and generate the unique force keys used to join the battle.
  - **Replay:** A link to the after-action replay, available once the scenario is complete.
- **Live monitoring & controls:** For the duration of the engagement, the administrator retains continuous visibility of tick progression and entity status for both belligerents, and may invoke the Fog of War control and toggle cartographic layers (zones, grid, boundaries, annotations) at any juncture, including mid-engagement.

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l. Player Command Interface

- **Join & status:** Force-key authentication for engagement entry, a continuously updated status summary (phase, residual time, own force composition), and inventory-cognisant tooling for formation deployment.

- **Command & control:** Designate a formation or discrete entity via the cartographic display or an enumerated list, assign its doctrine, plot or rescind a waypoint route, transmit orders, and transition to spectator mode for passive observation.
- **Battle control:** Designate the force as Ready, initiate the engagement once readiness is attained bilaterally, progress ticks incrementally or via automated advancement at a configurable rate, and invoke Play Again to recommence the identical engagement upon its conclusion.
- **Situational displays:** An enumerated roster of deployed entities with continuously updated health, ordnance and fuel status; own and hostile HVT condition; territorial-incursion notifications; a Win / Loss / Stalemate determination banner; a results overlay; a pursuit-view spectator camera; and a timeline control for retrospective tick examination absent interruption to the ongoing engagement.

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m. 3D Visualisation & Map Tools

- **Units & effects:** Terrain-conformant three-dimensional vehicular representations bearing continuously updated health and morale indicators, nominal labelling, tracer and impact weapon-effect rendering, and distinguishable HVT markers per belligerent.
- **Routes & zones:** Formation routes rendered as sequential waypoint traces, and BLUE/RED deployment zones depicted as unambiguously labelled, force-differentiated areas upon the cartographic surface.
- **Navigation:** A 2D/3D display toggle, zoom, rotational and tilt controls, a locational search facility, an adaptive coordinate grid responsive to zoom magnitude, and terrain-precise pointer resolution ensuring cursor input registers exactly at the intended ground reference.

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n. After Action Review — Replay & Results Dashboard

- **Full battle replay:** Every persisted tick is reconstructible from inception to conclusion, employing the identical three-dimensional representations and fire-effect rendering observed contemporaneously, via a comprehensive timeline control offering variable playback velocity, incremental step-forward/back navigation and preconfigured camera vantages.
- **Kill feed:** A granular, tick-resolved ledger establishing attacker-victim attribution — the originating force and entity, the affected force and entity, the ordnance or round expended, and the precise tick of occurrence — such that every casualty remains fully auditable.
- **Results dashboard with AI Briefing:** An outcome determination banner (BLUE / RED / Stalemate) with accompanying rationale; key-performance indicator panels enumerating entities deployed, attritted and surviving, together with kills attributable to each belligerent; and analytical visualisations encompassing force-strength trajectory over time, a comparative force assessment, attrition by unit classification, HVT health trajectory across the engagement, and a kill-ratio decomposition.

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o. Force Hierarchy (Order of Battle Register)

- **Structure:** An autonomous order-of-battle register — hierarchically organised as Lands, thence Forces (by typology), thence Units, thence SubUnits — sustained independently of any particular scenario's deployment configuration.
- **Management:** Aggregate enumeration at each hierarchical tier, a view-selection mechanism spanning Forces / Units / SubUnits, and provision for addition, amendment and removal, with nomenclatural duplication under a common parent rejected so as to preserve registral integrity.

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p. Access Control & Roles

- **Administrator:** A staff-level authentication credential conferring access to the Command Panel, scenario configuration, the asset catalogue, the Force Hierarchy register, replay functionality and results.
- **Force Player:** Access is circumscribed to a singular force and a singular scenario by means of a unique force key. A participant may command and deploy solely their own force, perceives only their own designated zone and inventory, and — with Fog of War enabled — perceives only those hostile entities currently within their own force's detection envelope.
- **Enforcement:** Every operative action is subject to server-side validation of role and permissible scope, with conventional web-request safeguards applied uniformly throughout.

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q. System Architecture & Interfaces

- **Backend:** An open-source web application framework, conjoined with a relational database, employed for the persistence of all scenario, entity, tick and results data.
- **Frontend:** Entirely browser-resident, utilising an open-source three-dimensional globe engine for terrain rendering and an open-source charting library for analytical presentation — obviating any requirement for client-side installation upon participant or administrator hardware.
- **Interfaces:** A documented application programming interface encompassing scenario administration, tick progression, entity and formation governance, objectives and zones, terrain and geography, analytics, and asset/force-hierarchy administration — thereby facilitating prospective interoperability with external systems.
- **Data model:** Structured, queryable records maintained for every scenario, force key, terrain grid, deployment zone, entity, HVT, simulation tick, casualty, platform/weapon asset, and the force hierarchy — underpinning comprehensive traceability from the live engagement through to after-action analysis.

Note: The entire solution — engine, 3D globe/terrain visualisation, database, charting/analytics, and any other library, framework or package used — must be built using only free and open-source software. No paid software, licensed packages, proprietary libraries, or components requiring a commercial subscription may be used at any stage of the PROJECT or the final deliverable.